THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** push or pull	Force.
2	**(c)** more inertia	Mass.
3	**(d)** 2 m/s squared	F/m.
4	**(c)** newton	N.
5	**(b)** third law	Action-reaction.
6	**(c)** 6 N	ma.
7	**(c)** inertia	Law of inertia.
8	**(a)** inertia	Inertia of rest.
9	**(c)** 100 N	mg.
10	**(d)** mass	Mass.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A body remains at rest or in uniform motion unless a force acts (2).	2
12	Product of mass and velocity (2).	2
13	Mass in kg, weight in N (1); Weight varies with g (1).	1+1
14	Force equals mass times acceleration (rate of change of momentum) (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Dust from carpet (1); Passengers falling backward (1); Coin on a card (1).	1+1+1
16	F = ma (1); 1000 x 3 (1); 3000 N (1).	1+1+1
17	Subtract (1); 10 - 4 (1); 6 N east (1).	1+1+1
	OR: Passenger moving forward when brakes are applied (1); Athlete running after finishing (1); Fruits falling when a tree is shaken (1).	1+1+1
18	p = mv (1); 5 x 4 (1); 20 kg m/s (1).	1+1+1
	OR: Add (1); 5 + 7 (1); 12 N (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	First law (1); Second law (1); Third law (1); Examples (1); Formulas (1).	1+1+1+1+1
	OR: Fuel burns (1); Gases expelled downward (1); Reaction pushes up (1); Third law (1); Space travel (1).	1+1+1+1+1
20	Definition (1); Rest (1); Motion (1); Example 1 (1); Example 2 (1).	1+1+1+1+1
	OR: Inertia (1); Seat belts (1); Airbags (1); Force reduced (1); Time increased (1).	1+1+1+1+1

Mistakes to avoid:

- Using mass in kg as weight in N (a 5 kg body weighs 50 N when $g = 10$), or saying that the mass of a body changes on the Moon.
- Using the applied force instead of the *net* force in $F = ma$ when friction acts.
- Saying that action and reaction cancel; they are equal and opposite but act on two different bodies.
- Forgetting to convert grams to kilograms (a 20 g bullet is 0.02 kg) in momentum and recoil problems.
- Dropping the negative sign (direction) of recoil velocity, or believing that a heavier gun recoils faster; recoil speed is inversely proportional to the gun's mass.
- Confusing inertia of rest with inertia of motion, or thinking that inertia depends on speed; it depends only on mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** backward	Inertia of rest.
2	**(d)** 10 kg m/s	mv.
3	**(c)** ma	Second law.
4	**(b)** forward	Inertia of motion.
5	**(d)** zero	Balanced.
6	**(a)** mv	Formula.
7	**(d)** friction	Rolling.
8	**(d)** acceleration	Change in motion.
9	**(c)** action and reaction	Third law.
10	**(b)** friction	Lubricant.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	1 N = force giving 1 m/s squared to 1 kg (2).	2
12	F = ma (1); = 8 N (1).	1+1
13	Every action has an equal and opposite reaction (2).	2
14	Force opposing motion between surfaces (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Rocket (1); Swimmer (1); Gun recoil (1).	1+1+1
16	W = mg (1); 50 x 10 (1); 500 N (1).	1+1+1
17	a = 3 m/s squared (1); v = at (1); 9 m/s (1).	1+1+1
	OR: Oil (1); Ball bearings (1); Polishing (1).	1+1+1
18	Quantity of matter vs force (1); kg vs N (1); Constant vs varies (1).	1+1+1
	OR: Reduce friction (1); Reduce wear (1); Reduce heating (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	p = mv (1); F = change in p/t (1); F = m(v - u)/t (1); F = ma (1); Units (1).	1+1+1+1+1
	OR: a = 2 m/s squared (1); F = 1600 N (1); Distance 100 m (1); Formulas (1); Units (1).	1+1+1+1+1
20	a = 5 (1); v = 25 (1); s = 62.5 (1); Units (1); Formulas (1).	1+1+1+1+1
	OR: Balanced (1); Unbalanced (1); Tug of war (1); Pushing a trolley (1); Effects (1).	1+1+1+1+1

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- Saying that action and reaction cancel; they are equal and opposite but act on two different bodies.
- Forgetting to convert grams to kilograms (a 20 g bullet is 0.02 kg) in momentum and recoil problems.
- Dropping the negative sign (direction) of recoil velocity, or believing that a heavier gun recoils faster; recoil speed is inversely proportional to the gun's mass.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** zero	No net force.
2	**(a)** mg	Formula.
3	**(b)** more friction	Roughness.
4	**(b)** towards the centre of the Earth	Down.
5	**(b)** 10 m/s squared	Gravity.
6	**(b)** does not change	Constant.
7	**(a)** friction	Friction.
8	**(d)** opposite to motion	Opposes.
9	**(b)** one-sixth of that on Earth	g/6.
10	**(a)** inertia	Prevent forward fall.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	W = mg (1); = 60 N (1).	1+1
12	Tendency of a body to resist change in its state (2).	2
13	Force of gravity on a body (1); W = mg (1).	1+1
14	Walking (1); Brakes (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Walking (1); Writing (1); Brakes (1).	1+1+1
16	Inertia of motion (1); Hold passengers (1); Prevent injuries (1).	1+1+1
17	Changes speed (1); Changes direction (1); Changes shape (1).	1+1+1
	OR: a = F/m (1); 50/10 (1); 5 m/s squared (1).	1+1+1
18	Friction (1); Air resistance (1); Unbalanced force slows it (1).	1+1+1
	OR: Larger inertia (1); Larger mass (1); F = ma (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Definition (1); Advantage 1 (1); Advantage 2 (1); Disadvantage 1 (1); Disadvantage 2 (1).	1+1+1+1+1
	OR: Card on glass (1); Coin on card (1); Flick (1); Coin drops (1); Inertia (1).	1+1+1+1+1
20	Mass (1); Weight (1); W = mg (1); 50 kg (1); 500 N (1).	1+1+1+1+1
	OR: Walking (1); Brakes (1); Wear (1); Heat (1); Writing (1).	1+1+1+1+1

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